Lecture 12

Partitions, branching and Young's orthogonal form

Objectives. *Lecture 11 proved that every weight of S_n is a content vector, the content vector of a standard Young tableau. This lecture proves the converse by counting: admissible transpositions link exactly the tableaux of one shape, and Lecture 7 gives $p(n)$ irreducibles. So partitions label the irreducibles, tableaux label their Gelfand–Tsetlin bases, and restriction removes one box. One vector then fixes every phase, and the matrices of the s_i follow: Young's orthogonal form.*

By the end of this lecture you will be able to:

- Decide from the axial distance whether $s_i T$ is standard, and link any two standard tableaux of one shape by admissible transpositions.
- Prove that the irreducibles of S_n are the V^λ, $\lambda \vdash n$, with $\dim V^\lambda = f^\lambda$ and, for $n \geq 2$, $\operatorname{Res}^{S_n}_{S_{n-1}} V^\lambda \cong \bigoplus_{\mu=\lambda-\square} V^\mu$.
- Prove Young's orthogonal form, and explain why the phases of an orthonormal GZ basis must be fixed together.
- Write the matrices of $s_1, \dots, s_{n-1}$ on a given V^λ, and compute character values from them.

12.1 Shapes and admissible transpositions

For $T \in \mathrm{SYT}(n)$, write

$$\alpha(T) = (c_1(T), \dots, c_n(T)).$$

By Proposition 11.14, every content vector of length n is $\alpha(T)$ for exactly one $T \in \mathrm{SYT}(n)$, and its *shape* is the shape of T.

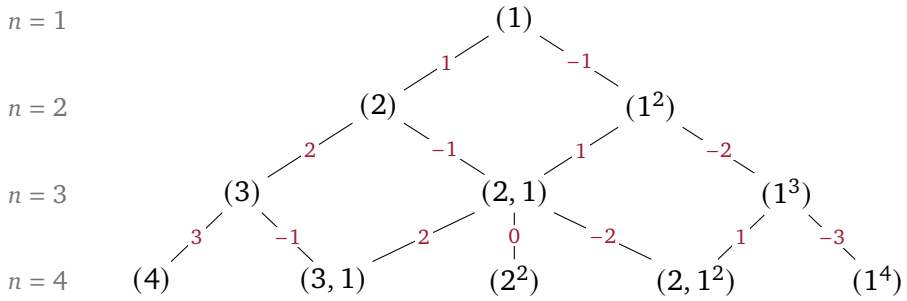
Definition 12.1 (Young's graph). *Young's graph* $\mathbb{Y}$ has as vertices the partitions of $1, 2, 3, \dots$, and an edge $\mu \to \lambda$ when the diagram of λ is that of μ with one box added: $\lambda = \mu + \square$ and $\mu = \lambda - \square$. The edge is labelled by the content of the added box.

A tableau $T \in \mathrm{SYT}(\lambda)$ is a path in $\mathbb{Y}$. By fact (a) of § 11.4, the boxes holding $1, \dots, k$ form a diagram, of shape $\lambda^{(k)}$ say, and

$$T \longmapsto \left(\lambda^{(1)} \to \lambda^{(2)} \to \cdots \to \lambda^{(n)} = \lambda\right).$$

The path starts at the box of 1, and step $k \geq 2$ adds the box of k, with label $c_k(T)$. Conversely, writing 1 in $(1,1)$ and k in the box added at step k turns a path from (1) to λ into a standard tableau, as each box comes after those above it and to its left. So $\mathrm{SYT}(\lambda)$ is the set of these

paths, and the labels along the path of T are $\alpha(T)$ without its first entry, 0. Up to $n = 4$:



This is Example 10.9's branching graph of S_4, with the same labels. Theorem 12.5 explains why.

Definition 12.2 (Axial distance)**.** For $T \in \mathrm{SYT}(n)$ and $1 \le i \le n-1$, the *axial distance* is $r = c_{i+1}(T) - c_i(T)$: Lecture 11's r for the vector $\alpha(T)$.

Lemma 12.3 (Admissible transpositions of tableaux)**.** *Let $T \in \mathrm{SYT}(n)$, $1 \le i \le n-1$, and r the axial distance.*

(i) If i and $i+1$ lie in one row of T, then $r = 1$; if they lie in one column, then $r = -1$. In both cases $s_i T$ is not standard.

(ii) Otherwise $s_i T$ is a standard tableau of the shape of T, with

$$\alpha(s_i T) = s_i \alpha(T),$$

and $r \neq \pm 1$.

So s_i is admissible for $\alpha(T)$ if and only if $s_i T$ is standard.

Proof. (i) A box between i and $i+1$ in their row would hold an entry strictly between them. So $i+1$ is the right-hand neighbour of i, and $r = 1$; in a column, $i+1$ lies just below i, and $r = -1$. In $s_i T$, $i+1$ then precedes i in that row or column.

(ii) *Standard.* Exchanging i and $i+1$ changes only the comparison of i with $i+1$, which share no row or column. The box of i in $s_i T$ is the box of $i+1$ in T, and conversely, so $\alpha(s_i T) = s_i \alpha(T)$.

$r \neq 1$. Let i lie in (p, q) and $i+1$ in (p', q'), with $p' \neq p$ and $q' \neq q$. If $r = 1$, then

$$q' - p' = q - p + 1.$$

If $p' < p$, then $q' < q$, so (p', q') lies above and to the left of (p, q), and by increase towards the lower right its entry $i+1$ is smaller than i. If $p' > p$, then $q' > q+1$, so $(p, q+1)$ lies above and to the left of (p', q'); it is a box, and its entry lies between i, on its left, and $i+1$. Both are impossible.

$r \neq -1$. The same, with rows and columns exchanged. ∎

12.2 Connectivity and the classification

Two content vectors are *linked* if a chain of transpositions, each admissible for the vector it is applied to, takes one to the other. As s_i is an involution, admissible for α if and only if for $s_i \alpha$, this is an equivalence relation.

Theorem 12.4 (Shapes are linked classes)**.** *Two content vectors of length n are linked if and only if they have the same shape.*

Proof. Only if. By Lemma 12.3(ii), an admissible s_i takes $\alpha(T)$ to $\alpha(s_iT)$, and s_iT has the shape of T.

If. For $\lambda \vdash n$, the *row tableau* T^λ fills the first row with $1, \dots, \lambda_1$, the second row with the next λ_2 numbers, and so on. By Lemma 12.3, it suffices to link each $T \in \mathrm{SYT}(\lambda)$ to T^λ by moves $T \mapsto s_iT$ with s_iT standard. By induction on n; $n = 1$ is clear.

Moving n. Let b be the last box of the last row of λ, which holds n in T^λ, and k the entry of T in b. For $j = k, k+1, \dots, n-1$ in turn, j is in b.

Then $j+1$ lies neither in the row of b, which ends at b, nor in its column, as nothing lies below b and the entries above it are smaller than j. So s_j moves $j+1$ into b and keeps the tableau standard (Lemma 12.3(ii)).

The rest. Now n is in b, and the other boxes form a standard tableau of shape $\lambda - b$. By induction it is linked to $T^{\lambda-b}$ by moves s_i with $i \le n-2$, which fix n. And $T^{\lambda-b}$ with n in b is T^λ. ∎

Theorem 12.5 (The classification). *Let $n \ge 1$.*

(i) $\mathrm{Spec}(n) = \mathrm{Cont}(n)$.

(ii) For each $\lambda \vdash n$ exactly one $V^\lambda \in \mathrm{Irr}(S_n)$ has as its weights the content vectors of shape λ, and $\lambda \mapsto V^\lambda$ is a bijection from the partitions of n onto $\mathrm{Irr}(S_n)$.

(iii) V^λ has a GZ basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$ with $X_k v_T = c_k(T)\, v_T$; so $\dim V^\lambda = f^\lambda$.

(iv) If $n \ge 2$, then

$$\mathrm{Res}^{S_n}_{S_{n-1}} V^\lambda \cong \bigoplus_{\mu=\lambda-\square} V^\mu,$$

Proof. The weights of one irreducible. Let $V \in \mathrm{Irr}(S_n)$. Its weights are content vectors (Theorem 11.12), and with a weight α they contain $s_i\alpha$ for every admissible s_i (after Definition 11.5). By Theorem 12.4, if they contain one content vector of shape λ, they contain all of them. Different irreducibles have no weight in common (Corollary 10.13).

Counting. So we may choose for each V a shape λ_V whose content vectors are weights of V, and $V \mapsto \lambda_V$ is one-to-one. By Theorem 7.9(ii) and Theorem 2.9,

$$\#\,\mathrm{Irr}(S_n) = \#\text{classes of } S_n = p(n).$$

So $V \mapsto \lambda_V$ is a bijection onto the partitions of n.

(i) and (ii). Every content vector is therefore a weight, which is (i). And V has no weight of a shape $\mu \ne \lambda_V$: those are weights of the V' with $\lambda_{V'} = \mu$. This is (ii), with V^λ the V with $\lambda_V = \lambda$.

The basis. By Corollary 10.13, $\gamma \mapsto \alpha(v_\gamma)$ maps the paths to V^λ one-to-one onto its weights, the $\alpha(T)$, $T \in \mathrm{SYT}(\lambda)$, one for each T (Proposition 11.14). Write v_T for the v_γ of weight $\alpha(T)$, and V_T for its line.

Branching: the last edge. Let $n \ge 2$, and let $\gamma = (U_1 \to \cdots \to U_n)$ be the path of v_T. By Proposition 10.8, a_k depends only on the edge $U_{k-1} \to U_k$, so the path $(U_1 \to \cdots \to U_{n-1})$ to U_{n-1} has the weight $(c_1(T), \dots, c_{n-1}(T))$. This is $\alpha(T')$, where T' is T with the box $\square$ of n removed, a standard tableau of shape $\lambda - \square$. By (ii) for S_{n-1}, $U_{n-1} = V^{\lambda-\square}$.

Branching: every box. If $\lambda - \square$ is a partition, some T holds n in $\square$: fill the rest by any standard tableau. So the edges into V^λ come from exactly the V^μ, $\mu = \lambda - \square$, each once (Theorem 10.3), which is (iv). ∎

So the branching graph of Definition 10.4 is Young's graph, with V^λ at λ, and by the branching step, repeated, the path of v_T is the path of T. On $\mathbf{1}$ each X_k acts by $k-1$, and on

sgn by $1-k$, the vectors of the one-row and the one-column tableaux; with § 11.5,

$$\mathbf{1} = V^{(n)}, \qquad \operatorname{sgn} = V^{(1^n)}, \qquad E = V^{(2,1)}.$$

12.3 The orthogonal form

Fix $\lambda \vdash n$ and an invariant inner product on V^λ (Theorem 5.9(iii)). The lines V_T are orthogonal (Lemma 11.1(ii)), so V^λ has orthonormal GZ bases, each vector fixed up to a scalar of modulus 1; one vector fixes them all.

Definition 12.6 (σ_T and inversions)**.** For $T \in \mathrm{SYT}(\lambda)$, $\sigma_T \in S_n$ sends the entry of T^λ in each box to that of T. An *inversion* of T is a pair $j < k$ with k in a higher row of T than j; $\operatorname{inv}(T)$ is their number.

Lemma 12.7 (Moves and inversions)**.** *Let $T \in \mathrm{SYT}(\lambda)$.*

(i) If s_iT is standard, then $\sigma_{s_iT} = s_i\sigma_T$, and $\operatorname{inv}(s_iT)$ is $\operatorname{inv}(T)+1$ if $i+1$ lies in a lower row of T than i, and $\operatorname{inv}(T)-1$ if in a higher one.

(ii) $\operatorname{inv}(T^\lambda) = 0$. If $T \neq T^\lambda$, then for some i, s_iT is standard and $\operatorname{inv}(s_iT) = \operatorname{inv}(T) - 1$.

Proof. (i) *The permutation.* In each box, the entry of s_iT is s_i of that of T, so $s_i\sigma_T$ of that of T^λ.

The inversions. By Lemma 12.3, i and $i+1$ lie in different rows. For $j \neq i, i+1$, $j < i$ if and only if $j < i+1$, so the pairs $\{j,i\}$ and $\{j,i+1\}$ exchange roles. Only the pair $\{i,i+1\}$ changes: it is an inversion of s_iT exactly when $i+1$ lies lower than i in T.

(ii) T^λ fills its rows in order. If no i has $i+1$ in a higher row than i, the rows of $1,2,\dots,n$ descend weakly, and $T = T^\lambda$. Otherwise $i+1$ is in neither the row nor the column of i, where it would lie above i; so s_iT is standard (Lemma 12.3), and the inversion-count formula applies. ∎

Theorem 12.8 (Young's orthogonal form)**.** *Let u_{T^λ} be a GZ vector in V_{T^λ}, and for $T \in \mathrm{SYT}(\lambda)$ let u_T be the component of $\sigma_T u_{T^\lambda}$ in V_T. Then $u_T \neq 0$, and $v_T = u_T/\|u_T\|$ defines an orthonormal GZ basis of V^λ such that:*

(i) For $T \in \mathrm{SYT}(\lambda)$ and $1 \le i \le n-1$, with r the axial distance,

$$s_i v_T = r^{-1} v_T + \sqrt{1 - r^{-2}}\, v_{s_iT},$$

the second term absent when s_iT is not standard, where $r = \pm 1$.

(ii) An orthonormal GZ basis $\{v'_T\}$ of V^λ satisfies (i) if and only if $v'_T = \theta v_T$ for every T, for one $\theta \in \mathbb{C}$ with $|\theta| = 1$.

Proof. The vectors u_T. By induction on $\operatorname{inv}(T)$: $u_T \neq 0$; $\sigma_T u_{T^\lambda} - u_T$ lies in the lines V_S with $\operatorname{inv}(S) < \operatorname{inv}(T)$; and whenever $T_1 = s_iT$ is standard with $\operatorname{inv}(T_1) = \operatorname{inv}(T) - 1$ and axial distance r_1,

$$\begin{aligned} u_T &= s_i u_{T_1} - r_1^{-1} u_{T_1}, \\ \|u_T\|^2 &= (1 - r_1^{-2}) \left\|u_{T_1}\right\|^2. \end{aligned} \tag{$*$}$$

If $\operatorname{inv}(T) = 0$, then $T = T^\lambda$ (Lemma 12.7(ii)) and $\sigma_T = e$.

The step. Let $\operatorname{inv}(T) = m \ge 1$, and $T_1 = s_iT$ any tableau as in $(*)$ (one exists, Lemma 12.7(ii)). By induction, $\sigma_{T_1} u_{T^\lambda} = u_{T_1} + x$, with $u_{T_1} \neq 0$ and x in the lines V_S

with $\operatorname{inv}(S) < m-1$. As $\sigma_T = s_i\sigma_{T_1}$ (Lemma 12.7(i)),

$$\sigma_T u_{T^\lambda} = s_i u_{T_1} + s_i x.$$

The first term. As s_iT_1 is standard, $r_1 \neq \pm 1$; by Theorem 11.2(ii), with $w \in V_T$,

$$s_i u_{T_1} = r_1^{-1} u_{T_1} + w,$$
$$\|w\|^2 = (1 - r_1^{-2}) \left\|u_{T_1}\right\|^2.$$

The rest. By Theorem 11.2 and Lemma 12.3, s_i maps each V_S into $V_S + V_{s_iS}$, or into V_S if s_iS is not standard, and $\operatorname{inv}(s_iS) < m$ when $\operatorname{inv}(S) < m-1$. So $u_T = w$, and the rest lies in lines V_S with $\operatorname{inv}(S) < m$.

(i) *Not standard.* If s_iT is not standard, then $r = \pm 1$, and $s_i v_T = r v_T$, which is $r^{-1} v_T$ (Lemma 12.3, Theorem 11.2(i)).

Upward. If $\operatorname{inv}(s_iT) = \operatorname{inv}(T) + 1$, then $(*)$ for s_iT, with $T_1 = T$, gives

$$s_i u_T = r^{-1} u_T + u_{s_iT},$$
$$\left\|u_{s_iT}\right\| = \sqrt{1 - r^{-2}}\, \|u_T\|;$$

divide by $\|u_T\|$.

Downward. If $\operatorname{inv}(s_iT) = \operatorname{inv}(T) - 1$, then $T_1 = s_iT$ has axial distance $-r$, and by $(*)$ and Theorem 11.2(ii), with $v = u_{T_1}$ and $w = u_T$,

$$s_i u_T = (1 - r^{-2})\, u_{T_1} + r^{-1} u_T;$$

divide by $\|u_T\| = \sqrt{1 - r^{-2}}\, \left\|u_{T_1}\right\|$.

(ii) Each v'_T is $\theta_T v_T$ with $|\theta_T| = 1$, and by (i) the coefficient of v'_{s_iT} in $s_i v'_T$ is

$$\theta_T \theta_{s_iT}^{-1} \sqrt{1 - r^{-2}}.$$

So (i) holds for $\{v'_T\}$ if and only if $\theta_{s_iT} = \theta_T$ whenever s_iT is standard; and any two tableaux of shape λ are joined by such moves (Theorem 12.4, Lemma 12.3). ∎

Example 12.9 (E in three bases; two guards). For $\lambda = (2,1)$, the tableau $T = s_2 T^\lambda$ has $\sigma_T = s_2$ and $\operatorname{inv}(T) = 1$. In E take for u_{T^λ} the vector $u = (1, 1, -2)$, of weight $(0, 1, -1)$ (§ 10.6). By Lecture 10's self-check,

$$s_2 u = -\frac{1}{2} u + \frac{3}{2} f_1,$$

so $u_T = \frac{3}{2} f_1$. In the *seminormal* basis (u, u_T), by Theorem 11.2(ii) with $r = -2$, s_2 acts by

$$\begin{pmatrix} -\frac{1}{2} & \frac{3}{4} \\ 1 & \frac{1}{2} \end{pmatrix},$$

whose first column has length $\sqrt{5}/2$: *the seminormal form is not orthogonal*. Normalising, with

$$\|u\|^2 = 6, \qquad \|u_T\|^2 = \frac{9}{2},$$

gives the basis (v_{T^λ}, v_T), in which s_2 acts by

$$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

Fixing lengths does not fix phases: in the orthonormal GZ basis $(v_{T^\lambda}, -v_T)$ both off-diagonal entries are

$$-\frac{\sqrt{3}}{2}.$$

12.4 Worked example

What are the matrices of S_4 on $V^{(3,1)}$? Its tableaux, content vectors and axial distances are

	T_1 	T_2 	T_3 
$\alpha(T)$	$(0,1,2,-1)$	$(0,1,-1,2)$	$(0,-1,1,2)$
$r,\ s_1$	1	1	-1
$r,\ s_2$	1	-2	2
$r,\ s_3$	-3	3	1

with $T_1 = T^{(3,1)}$, $T_3 = s_2 T_2$ and $T_2 = s_3 T_1$. As

$$\sqrt{1-\frac{1}{4}} = \frac{\sqrt{3}}{2}, \qquad \sqrt{1-\frac{1}{9}} = \frac{2\sqrt{2}}{3},$$

Theorem 12.8 gives, in the basis $(v_{T_1}, v_{T_2}, v_{T_3})$,

$$D(s_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

$$D(s_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix},$$

$$D(s_3) = \begin{pmatrix} -\frac{1}{3} & \frac{2\sqrt{2}}{3} & 0 \\ \frac{2\sqrt{2}}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Branching. $D(s_1)$ and $D(s_2)$ fix v_{T_1}, and on (v_{T_2}, v_{T_3}) they are the matrices of $V^{(2,1)}$, as removing 4 from T_2 and T_3 leaves the two tableaux of shape $(2,1)$:

$$\operatorname{Res}^{S_4}_{S_3} V^{(3,1)} \cong V^{(3)} \oplus V^{(2,1)},$$

as the branching rule of Theorem 12.5 says.

A Coxeter relation. Multiplying out, both $D(s_2)D(s_3)D(s_2)$ and $D(s_3)D(s_2)D(s_3)$ equal

$$\begin{pmatrix} -\frac{1}{3} & -\frac{\sqrt{2}}{3} & \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{2}}{3} & \frac{5}{6} & \frac{\sqrt{3}}{6} \\ \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{6} & \frac{1}{2} \end{pmatrix},$$

the matrix of $(2\,4)$. Its trace is 1.

The character. So

$$\chi^{(3,1)}\big((1\,2)\big) = \operatorname{tr} D(s_1) = 1,$$

as for $(2\,4)$. In § 6.6's table the irreducibles of degree 3 are W, with $\chi_W((1\,2)) = 1$, and $\operatorname{sgn} \otimes W$, with -1. So $V^{(3,1)} \cong W$, as Lecture 8 promised.

Self-check. Find the matrices of s_1, s_2, s_3 on $V^{(2,2)}$, and $\chi^{(2,2)}((1\,2))$. *Answer:*

	T $\begin{array}{\|c\|c\|}\hline 1&2\\\hline 3&4\\\hline\end{array}$	$T' = s_2T$ $\begin{array}{\|c\|c\|}\hline 1&3\\\hline 2&4\\\hline\end{array}$
$\alpha(T)$	$(0,1,-1,0)$	$(0,-1,1,0)$
$r,\ s_1$	1	-1
$r,\ s_2$	-2	2
$r,\ s_3$	1	-1
$s_1v,\ s_3v$	v_T	$-v_{T'}$

and $D(s_2)$ is Example 12.9's orthogonal matrix. The character value is the trace,

$$\chi^{(2,2)}\big((1\,2)\big) = 1 - 1 = 0,$$

as for $\chi_E \circ \pi$ in § 6.6; and $D(s_1s_3)$ is the identity, as s_1s_3 lies in V_4, the kernel of $\rho_E \circ \pi$.

Summary.

- *The main results:* the tableaux of one shape are linked (Theorem 12.4); $\mathrm{Spec}(n) = \mathrm{Cont}(n)$, the irreducibles V^λ and branching (Theorem 12.5); Young's orthogonal form (Theorem 12.8).
- *Handed on:* the classification, branching and Young's graph to Lectures 13 and 14.
- *Next:* Lecture 13 finds all the characters χ^λ at once.

References

Vershik and Okounkov, *A new approach to the representation theory of the symmetric groups. II* (arXiv:math/0503040): Young's graph, pp. 19–20; Prop. 5.3 and Lemma 5.4, p. 20; Cor. 5.5, p. 21; Thm 5.8, p. 22; §6, pp. 22–23. Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Thm 3.3.7, p. 106; Prop. 3.3.9 and Cor. 3.3.11, p. 107; §3.4.1–3.4.2, pp. 110–113.[1]

[1] They write $\mathrm{Tab}(\lambda)$ for $\mathrm{SYT}(\lambda)$ and S^λ for V^λ, and order the tableaux by the length of σ_T as a word in the s_i (VO p. 22, CST Prop. 3.4.1).